

Flask Day-1

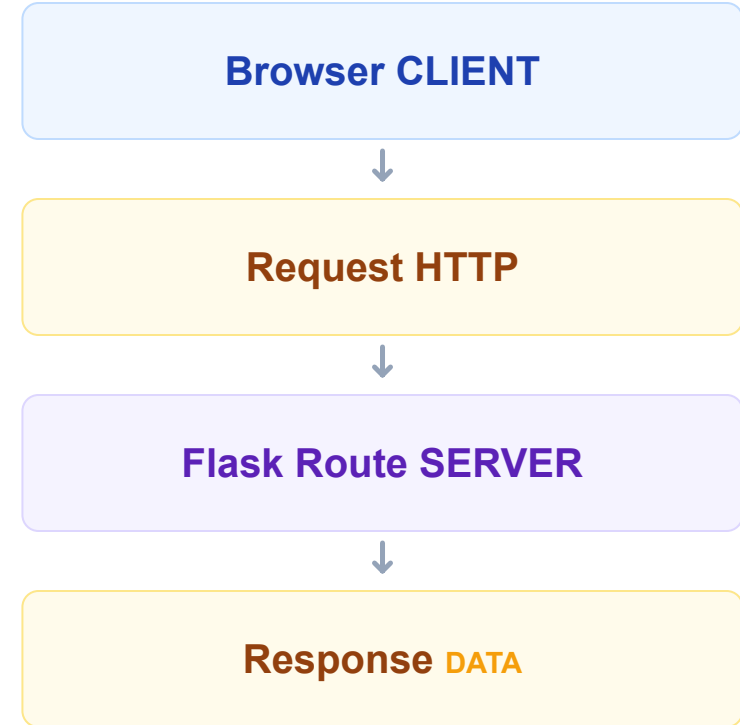
Class Presentation

Introduction to Web Development & Flask

Trainer

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Request-Response Cycle



What students must master today

The exam repeats core ideas with different wording.

01

Web App vs Website

Know interaction vs static reading

02

Client - Server

Browser asks, server responds

03

Browser & Web Server

Chrome is client; Nginx/Gunicorn serve apps

04

HTTP/HTTPS

Request-response protocol; HTTPS is secure

05

Flask Route

URL is mapped to a Python function

06

Trailing Slash

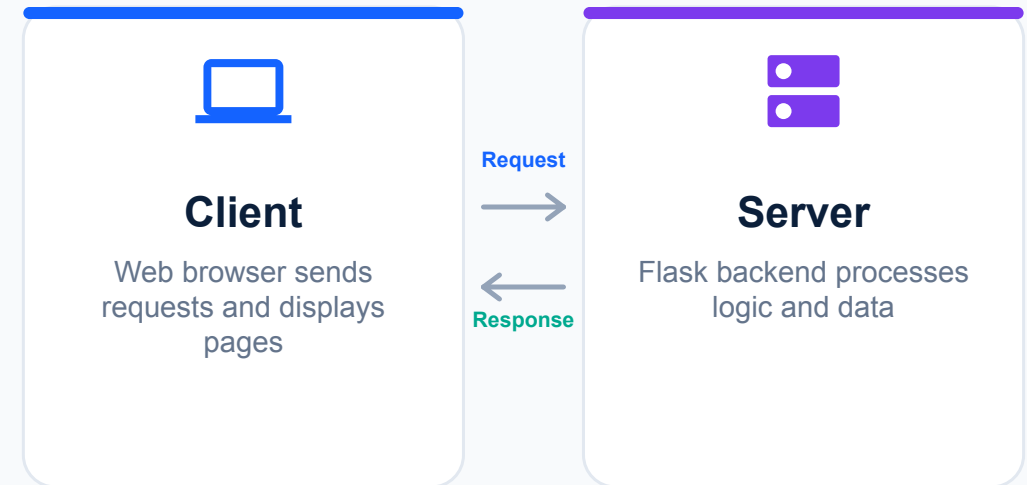
Strict matching behavior in URL paths

SECTION 01

Part 1: Web Basics

The Mental Model

Build the mental model before writing Flask code.



Website vs Web Application

Students should identify “read only” vs “interactive”.

Website

- Mostly displays information
- Mostly read-only
- Usually no login required

Example

Company profile page

Web Application

- User can interact and change data
- Read + write + process
- Often has login, forms, dashboard

Examples

Gmail, LMS, Amazon

Memory Line

Website = read info | Web app = do something

Class Question

**Is Codegnan LMS a website or web application?
Why?**

Expected Answer

Web application, because students login, watch classes, submit work and receive responses.

Full Stack Development: The 4 Layers

Connect each layer to a technology students already know.



Frontend



HTML, CSS, JS, React

What user sees

Backend



Python, Flask

Logic and processing

Database



MySQL

Stores data

Deployment

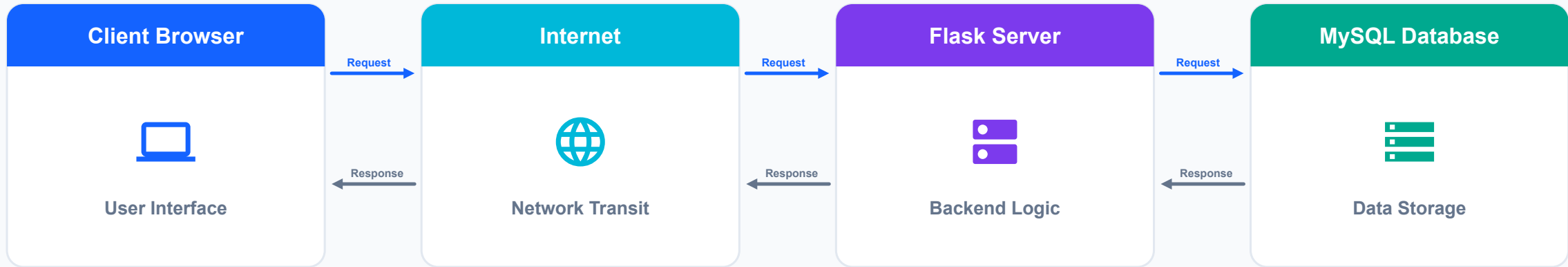


AWS, Nginx, Gunicorn

Makes app public

Client-Server Architecture

Most web apps follow this request-response relationship.



How the Request-Response Loop Works

- **Client requests service or data:** The user interacts with the browser, which sends a structured HTTP request over the Internet.
- **Server processes and responds:** The Flask backend server logic handles processing, manages security, and routes the returned data.
- **Database storage:** MySQL acts as the persistent database server to store and fetch dynamic application information.

Key Concept

Response Comes Back

Every web action is an absolute loop. A client browser cannot display dynamic updates without initiating a request first.

Once the server processes the data, the loop completes as the server's response brings content back to the user's screen.

Browser, Internet and Web Server

Three terms students must not confuse.

Browser

Meaning:

Client-side application that opens websites, renders HTML and executes JavaScript

Examples / Clues:

Chrome, Firefox, Safari, Edge

Internet

Meaning:

Global network that connects computers and servers

Examples / Clues:

Used to reach remote apps

Web Server

Meaning:

Hosts application and responds through URL or IP address

Examples / Clues:

Flask dev server, Nginx, Apache

Chrome

Browser

Nginx

Web server

EC2

Remote server

Gunicorn

Runs Flask app

HTTP vs HTTPS

HTTPS is HTTP with security through encryption.

HTTP (Unsecured)

ENCRYPTION

Not encrypted

DEFAULT PORT

Port 80

BEST SUITED FOR

Learning and local offline demos

SECURITY RISK

Data can be intercepted and exposed

HTTPS (Secured)

ENCRYPTION

Encrypted via SSL/TLS

DEFAULT PORT

Port 443

BEST SUITED FOR

Preferred for all real public websites

SECURITY LEVEL

Data is fully protected in transit

Exam Memory Trick

HTTPS

S = Secure

443 = Secure web port

SECTION

Part 2: Flask Fundamentals

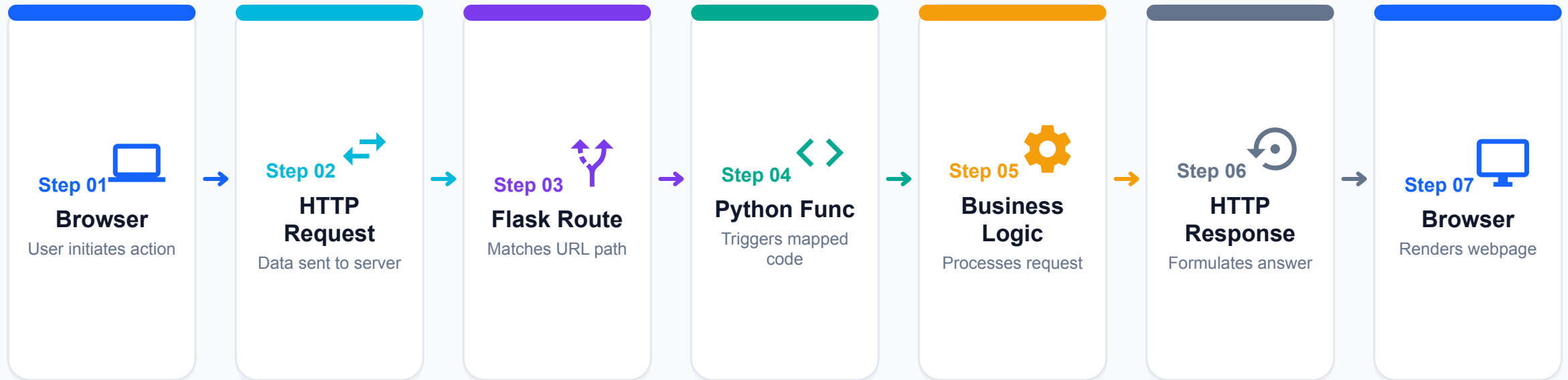
Translate web concepts into Flask code.

```
from flask import Flask
app = Flask(__name__)

@app.route("/")
def index():
    return "Web Concept → Flask Code"
```

Request-Response Cycle in Flask

The browser asks. Flask answers.



What is Flask?

A lightweight Python web framework used to build web applications and APIs.



Flask helps Python handle web requests.

- **Creates web applications** using Python.
- **Maps URLs** dynamically to Python functions.
- **Flexible outputs:** can return text, HTML, JSON APIs, or files.
- **Lightweight design:** works perfectly for small-to-medium apps.

Philosophy

"Micro" does not mean weak.

It means Flask provides the essential core features and grants developers the freedom to choose their own database, authentication, and extra libraries.

Common Applications

Websites • **Dashboards** • **LMS Modules** • **REST APIs** • **Admin Panels**

Minimal Flask Application

Students must understand every line.



```
1 from flask import Flask
2
3 app = Flask(__name__)
4
5 @app.route("/")
6 def home():
7     return "Hello World"
8
9 if __name__ == "__main__":
10     app.run(debug=True)
```

import Flask

Bring Flask class into our program

Flask(__name__)

Create the Flask app object

@app.route("/")

Map root URL to next function

return

Send response to browser

app.run()

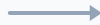
Start local development server

Route Decorator: URL -> Function

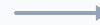
This is the heart of Flask Day-1.



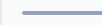
1. Request URL
/about



2. Decorator
@app.route()



3. Python Function
def about()



4. HTML Response
"About Codegnan"

Code Example

```
1@app.route("/about")
2def about():
3    return "About Codegnan"
```

Student Explanation

When the user opens **/about**, Flask locates the route decorator and executes the python function directly beneath it.

Trailing Slash Rules - Most Asked Topic

Spend extra time here. Many exam questions test this exact concept.

```
@app.route("/about/")
```

Directory-like canonical path

Both URLs work. Accessing `/about` automatically redirects to `/about/` with a trailing slash.

Slash at end = Directory = Redirect

```
@app.route("/about")
```

File-like exact path

Strict matching. Only `/about` works. Accessing with a trailing slash results in a 404 error.

No slash = Exact only

Class Drill & Practical Scenarios

- If defined route is `/profile/` and user opens `/profile` → Flask redirects automatically to `/profile/`.
- If defined route is `/profile` and user opens `/profile/` → It will **not match** by default (404 Page Not Found).

Local Server vs Production Deployment

Students should know why localhost is not enough.

Local Development Server

localhost / 127.0.0.1

- Used for building and testing
- Usually private to developer machine



Remote Server with Static IP

AWS EC2 / cloud server

- Connected to internet
- Accessible publicly through URL/IP



Production Service Auto Start

systemctl enable

- Starts service automatically
- Ensures app runs when server reboots

Production Stack

Browser→Nginx→Gunicorn→Flask→MySQL

Quick Class Quiz

Ask students before closing the session.

Flask Day 1



1 What does `@app.route("/")` do?

ANSWER: Maps root URL to a function

2 If route is `/contact/`, what happens when user opens `/contact`?

ANSWER: Redirects to `/contact/`

3 What is Chrome in web architecture?

ANSWER: Browser / client-side app

4 Why deploy to a remote server?

ANSWER: To make app publicly accessible

5 What does `systemctl enable` do?

ANSWER: Starts service automatically on boot



Teaching close: “Do not memorize options. Understand the clue words.”

Thank You!

Python Full Stack Development Course Completion



Keep in Touch

Practice well! If you have any questions or need guidance, feel free to reach out anytime.

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 **Rule of Thumb:** Focus on concepts, not option memorization.